Huawei Access Node Hebbal Edge Routing Protocol Flap Recovery (ALARM_1805)

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1805

Description

Hebbal Edge Access Node HEB-ACC-40, running Huawei access node software, exhibits a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on the Access Role, precipitating a Warning Severity Alarm Code ALARM_1805. This phenomenon is a manifestation of a complex interplay between the BGP, OSPF, and ISIS routing protocols, as well as the underlying hardware mechanics of the Huawei Access Node. Specifically, the occurrence of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on the Access Node indicates a disruption in the normal functioning of the routing protocols, which in turn affects the node's ability to forward traffic. This may be due to various factors, including misconfiguration, hardware issues, or changes in the network topology. To mitigate this issue, it is essential to identify and address the root cause of the routing protocol flap, ensuring that the Access Node recovers from the failure and resumes normal operations.

Pre-requisite Checks

- Verify the power and connectivity of the Hebbal Edge Access Node HEB-ACC-40.

- Check the status of the redundant links to ensure they are operational and not contributing to the routing protocol flap.

- Confirm that the Huawei access node software is up-to-date and running the latest version.

- Gather relevant logs and configuration files for further analysis.

Potential Root Causes

Cause	Probability
Misconfigured BGP peer	High
Hardware issue with the access node's routing engine	Medium
Changes in the network topology affecting the access node's routing protocols	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp peer, Expected Output: BGP Peer Status
------------------------ Peer IP Address State Description
------------------------ --------- --------------- 10.10.10.10
Established Established peer, Description: This command displays the
status of BGP peers on the access node, helping to identify any
misconfigured peers contributing to the routing protocol flap.
```

```
Command: display ospf neighbor, Expected Output: OSPF Neighbor Status
------------------------ Neighbor IP Address State Description
------------------------ --------- --------------- 10.10.10.10 Full
Established neighbor, Description: This command displays the status of
OSPF neighbors on the access node, aiding in the identification of any
adjacency loss issues contributing to the routing protocol flap.

Command: display isis lsp, Expected Output: ISIS LSP Status
------------------------ LSP ID State Description
------------------------ --------- ---------------
10.10.10.10:10.10.10.10 Up Established LSP, Description: This command
displays the status of ISIS LSPs on the access node, assisting in the
identification of any flooding issues contributing to the routing
protocol flap.
```

Resolution Steps

Step 1: Step: Verify and correct BGP peer configuration, Command: configure terminal router bgp 10 neighbor 10.10.10.10 remote-as 10 neighbor 10.10.10.10 update-source Loopback0, Description: This step involves verifying the BGP peer configuration on the access node and correcting any misconfigurations that may be contributing to the routing protocol flap.

Step 2: Step: Reset the access node's routing engine, Command: system-view reset routing-engine, Description: This step involves resetting the access node's routing engine to resolve any hardware issues that may be contributing to the routing protocol flap.

Step 3: Step: Re-run OSPF neighbor establishment, Command: ospf network-type broadcast ospf process-id 10 ospf neighbor 10.10.10.10, Description: This step involves re-running OSPF neighbor establishment on the access node to resolve any adjacency loss issues contributing to the routing protocol flap.

Post-Check Verification

```
Command: display bgp peer, Expected Output: BGP Peer Status
------------------------ Peer IP Address State Description
------------------------ --------- --------------- 10.10.10.10
Established Established peer, Description: This command verifies that
the BGP peer is established and not contributing to the routing
protocol flap.

Command: display ospf neighbor, Expected Output: OSPF Neighbor Status
------------------------ Neighbor IP Address State Description
------------------------ --------- --------------- 10.10.10.10 Full
Established neighbor, Description: This command verifies that the OSPF
neighbor is established and not contributing to the routing protocol
flap.

Command: display isis lsp, Expected Output: ISIS LSP Status
------------------------ LSP ID State Description
------------------------ --------- ---------------
10.10.10.10:10.10.10.10 Up Established LSP, Description: This command
verifies that the ISIS LSP is established and not contributing to the
routing protocol flap.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: configure terminal undo router bgp 10, Description: This step involves restoring the original BGP configuration to resolve any issues introduced during the resolution process.

- Step: Reset the access node's routing engine, Command: system-view reset routing-engine, Description: This step involves resetting the access node's routing engine to resolve any issues introduced during the resolution process.

- Step: Re-run OSPF neighbor establishment, Command: ospf network-type broadcast ospf process-id 10 ospf neighbor 10.10.10.10, Description: This step involves re-running OSPF neighbor establishment on the access node to resolve any issues introduced during the resolution process.